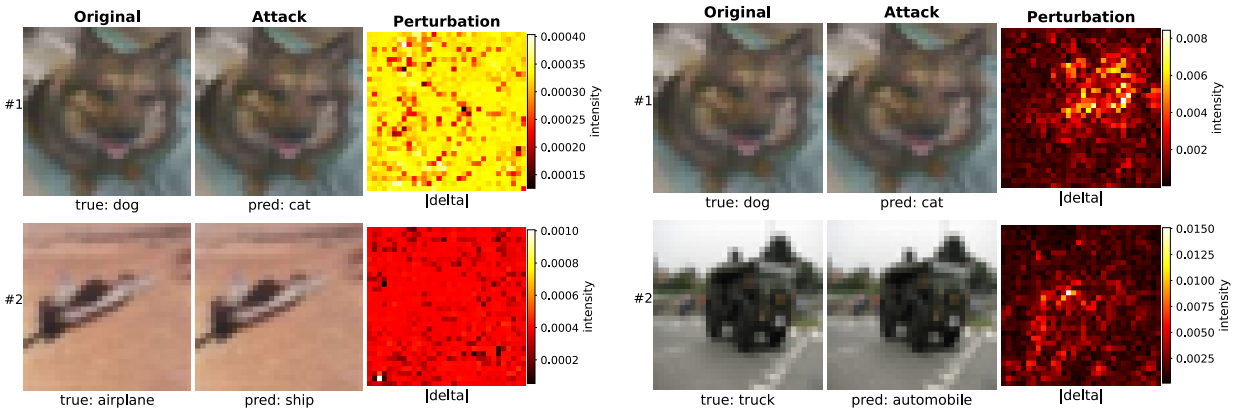
Table 1: **ML Application: Non-Uniform Adversarial Attack**: Attack performance under weighted ℓ_p perturbation constraints on CIFAR-10. Perturbations are constrained to $\Delta(x) = \{\delta \in \mathbb{R}^d : \|W\delta\|_p \leq \epsilon_p,\ 0 \leq x + \delta \leq 1\}$, where W encodes channel-spatial correlations. Evaluated on $N = 500$ samples with step size $\alpha = 10^{-3}$ over 500 iterations.

p	Method	Adv. Acc. (%) ↓	Success (%) ↑	Time/iter (s)
0.5	PGD	98.0	2.0	0.042
	Hom-PGD$^+$	**85.4**	**14.6**	0.042
1	PGD	97.2	2.8	0.041
	Hom-PGD$^+$	**83.6**	**16.4**	0.042
2	PGD	79.4	20.6	0.041
	Hom-PGD$^+$	80.0	20.0	0.042
4	PGD	78.6	21.4	0.041
	Hom-PGD$^+$	79.2	20.8	0.042



(a) PGD

(b) Hom-PGD

Figure 1: Successful adversarial examples under weighted $\ell_{1/2}$ constraints. Hom-PGD$^+$ achieves 7× higher success rate for non-convex regions ($p \leq 1$) with negligible overhead, as it reparameterizes the feasible region into a unit ball, avoiding the convergence issues of projecting onto non-convex sets.

Table 2: **Additional Baselines for the Chance Constrained Problems**: JCC-OPF baseline results (200-bus system, 100 scenarios). We compare two convex approximations for the chance constraint $\mathbb{P}[g(x, \xi) \leq 0] \geq 1 - \epsilon$: (i) Robust Scenario, enforcing $g(x, \xi^{(i)}) \leq 0$ for all scenarios; (ii) CVaR Approximation. Both are solved using MOSEK. **Scalability limitation:** For 1,000 scenarios or larger systems, MOSEK fails, motivating scalable iterative methods such as ours.

Solver	Status	Time (s)	Objective	Feas. Rate
CVaR	optimal	73.7	0.683	1.00
Robust	optimal	74.0	0.684	1.00

Table 3: **Extended Scaling Experiments on QCQP**: QCQP scaling experiments ($n_{\text{var}} = 100$). Problem complexity scales as $O(n_{\text{qua}} \cdot n_{\text{var}}^2)$. **Key finding:** Hom-PGD$^+$ runtime remains nearly constant as constraint count grows, achieving $>20\times$ speedup at 1000 constraints while maintaining comparable objective quality.

n_{qua}	Hom-PGD$^+$ Obj	IPOPT Obj	Hom-PGD$^+$ (s)	IPOPT (s)	Speedup
10	-0.470	-0.483	12.0	2.3	$0.2\times$
100	-0.455	-0.499	12.1	25.8	$\mathbf{2.1}\times$
1000	-0.453	-0.430	12.3	255.6	$\mathbf{20.7}\times$